CSC17104 – PROGRAMMING FOR DATA SCIENCE
FINAL PROJECT

1. Introduction

In this project, your group of **2–3 students** carries out a complete data science workflow on a real-world dataset of your choice: from finding raw data to answering meaningful questions and communicating what you found. It is also where you show that you can use the course tools (Linux command line, Git & GitHub, Conda, Jupyter, Python, NumPy, pandas, Matplotlib, scikit-learn) to work **reproducibly and as a team**.

Your group will:

- Find a public, real-world dataset that interests the team, and get it approved.
- Explore and preprocess the data, documenting every decision.
- Formulate **2 × n questions** (n = number of members), including **at least one answered with a machine learning model**, that meet the depth requirements in Section 4.3.
- Answer each question with analysis, visualization and evidence, then present your results.

2. Dataset requirements

- **Real-world and public:** collected from actual observations, measurements or events. Synthetic datasets are not allowed. Possible sources: Kaggle, UCI Machine Learning Repository, Data.gov, Google Dataset Search, Vietnamese open data portals, or other reputable sources.
- **Size and format:** at least ~3,000 rows and ~10 columns, in CSV, Excel, JSON or a similar structured format. Combining several sources is allowed.
- **Licensed and traceable:** licensed for educational use, and you must be able to say who collected the data and how.
- **Not too clean, not overused:** avoid toy datasets such as Iris or Titanic.
- **Declare existing work:** if public notebooks or analyses of your dataset exist (e.g. on Kaggle), list them in your README and explain how your questions differ. Reusing their questions or code without attribution is plagiarism.

3. Project workflow

Follow the five phases below. Phases 4.2–4.4 are iterative: you will often go back to earlier phases as you understand your data better. The checklists are a reference framework, not a rigid list: adapt them to your data, skip what is irrelevant, and add what your data needs.

3.1. Data collection

Answer the following in your notebook:

- **Subject:** what topic does the data cover, and what real-world context does it represent?
- **Source:** platform and full URL, original author or organization, publication date.
- **License:** which license applies, and are there restrictions or attribution requirements?
- **Collection method:** how the data was collected (survey, sensors, administrative records, scraping…), from which population, over what period, and with what known limitations or biases.
- **Motivation:** why your group chose this dataset.

3.2. Data exploration

Investigate the structure, quality and characteristics of your data. **Go deep on the columns relevant to your questions**, and summarise the rest briefly instead of producing one chart per column.

Aspect	What to check
Overview	Number of rows and columns; what one row represents; meaning and data type of each column; columns to drop or convert; duplicated or empty rows, and whether to remove them (justify).
Numerical columns	Distribution shape, centre and spread; whether min/max are plausible; outliers (IQR, z-score) and whether they are genuine or errors; impossible or placeholder values (negative ages, 999, -1).
Categorical columns	Number of distinct values and the most frequent ones; balance; inconsistent labels ("Male", "male", "M") and typos; rare categories that should be grouped.
Missing data	Count and percentage per column; whether missingness follows a pattern; why values are missing and how you will handle each column (drop, impute, separate category).
Relationships	Correlations between numerical variables; cross-tabulations and grouped statistics for important combinations; surprising relationships.

End the section with a short summary: 3–5 key observations, serious data quality issues, your preprocessing plan, and the patterns that could lead to questions.

3.3. Question formulation

Formulate **2 × n questions** (a group of 3 needs 6). Each question has **one owner**, and each member owns two questions. **At least one question must be answered with a machine learning model.** For each question, state:

- **The question**, precise enough to be answered with your data.

- **Motivation:** why it is worth answering, who would care about the answer (e.g. buyers, city planners, researchers), and what decision it informs.

Depth requirements for non-ML questions

Each non-ML question must clearly use **at least two** of the criteria below, and your questions together must cover **at least four** of the six.

	Criterion	What it means
D1	**Combining data**	Merging with another table or a second dataset. Verify the merge: row counts before and after, unmatched keys.
D2	**Reshaping & time**	Pivot or melt, resampling, rolling windows, lags, period-over-period comparisons.
D3	**Derived variables**	New, justified variables built from raw columns (e.g. price per m^2, distance to the nearest metro station).
D4	**Controlling for a confounder**	Checking whether a pattern still holds within subgroups, or after accounting for a third variable.
D5	**Quantifying uncertainty**	Confidence intervals (e.g. bootstrap), statistical tests or effect sizes, not just comparing two averages.
D6	**Robustness check**	Showing whether the conclusion changes under another reasonable choice: cleaning rule, outlier threshold, time window.

Not substantial: a question answered by one `groupby`, `value_counts` or `describe` plus one chart; describing a single variable (that belongs in exploration); a question with an obvious answer, such as "Do bigger houses cost more?"

- *Weak:* Which district has the highest average house price?
- *Strong:* After accounting for floor area and legal status, which districts are over- or under-priced per m^2, and did prices near the new metro line change after it opened? (D1–D5)

Requirements for the ML question

The model is a means to answer the question, not the goal. Your ML question must:

1. Define a prediction task with a clear user, using only features available at prediction time. List the columns you excluded to avoid **data leakage**.
2. Use an appropriate split: **time-based** for temporal data, **group-based** when several rows belong to the same entity (user, store, patient).
3. Compare **at least two models and a trivial baseline** (majority class, mean, "same as last period"), with all preprocessing inside a scikit-learn `Pipeline`.
4. Tune with cross-validation on the training set only; evaluate on the test set once, with a **metric you justify** (e.g. why accuracy misleads on imbalanced classes).
5. **Analyse errors:** where does the model fail (subgroups, value ranges, example rows)?

6. **Interpret the model** (e.g. permutation importance) and answer the original question, not just report a score.

3.4. Data analysis

For each question:

- **Preprocessing:** describe the steps in Markdown so a reader understands the workflow without reading the code, then implement them. After every step that changes the data (filtering, merging, dropping), **check and report the row count** and explain any change.
- **Analysis:** explain which method you use and why it fits the question, then implement it.
- **Results:** at least two well-labelled plots (title, axes, units, legend). Answer the question explicitly with specific numbers, and discuss practical meaning, surprises and limitations.

3.5. Summary and reflection

- **Key findings:** the 3–5 most important insights, highlighting the most surprising one.
- **Limitations** of the data, the methods and the scope.
- **Future work:** the questions, data or methods you would add with more time.
- **Individual reflection** (each member): the difficulties you faced and how you overcame them, what you learned, and how you used LLM tools.

4. Deliverables and repository structure

Your final repository should follow this structure:

```
project-name/
├── README.md
├── environment.yml
├── LLM_USAGE.md
├── data/raw/, data/processed/  # large files are not committed
├── src/                        # reusable Python modules
├── notebooks/
│   ├── 01_collection_exploration.ipynb
│   ├── 02_analysis.ipynb        # split per question if large
│   └── 03_ml_question.ipynb
└── slides.pdf
```

The **README.md** must contain: team members and student IDs; question ownership and task breakdown; dataset description, source, license and related public notebooks; the list of questions; key findings; and how to run the project (create the environment, download the data, run the notebooks in order).

5. Presentation and individual Q&A

- Each group has [15] minutes to present, followed by [10] minutes of Q&A.

- Tell the story of your data: the questions, the key findings, and how you know they are right. Do not walk through code line by line.
- Every member presents the questions they own.
- In the Q&A, the teacher may ask any member about any part of the project, including code they did not write. Individual grades may differ from the group grade, based on the Q&A and the Git history.